REVISED SEPTEMBER 2026

Toyota Techstream Imaging Guide

Vehicle Control History · PCS Freeze Frame Data · Forward-Facing Camera Images

A step-by-step field procedure for retrieving and preserving Toyota and Lexus Vehicle Control History (VCH) and Pre-Collision System (PCS) data using TIS Techstream, ending with a checklist of every file that must be in hand before leaving the inspection.

Compiled from the source material listed on the final page.
For use by trained crash investigators with lawful authority to retrieve vehicle data.

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Source Material

General Precautions

Before Connecting to the Vehicle

Confirm that the vehicle's battery, ground cables, and positive 12-volt cables are properly connected and undamaged. Inspect the airbag ECU and its wiring, and the DLC itself, for crash damage that may complicate or corrupt a retrieval.

Photograph the vehicle, the VIN plate, the odometer, and the DLC before you connect anything. Record the date and time from your own watch alongside the laptop clock: Techstream timestamps the retrieval file from the PC clock, and VCH records carry their own clock reference that must be reconciled later.

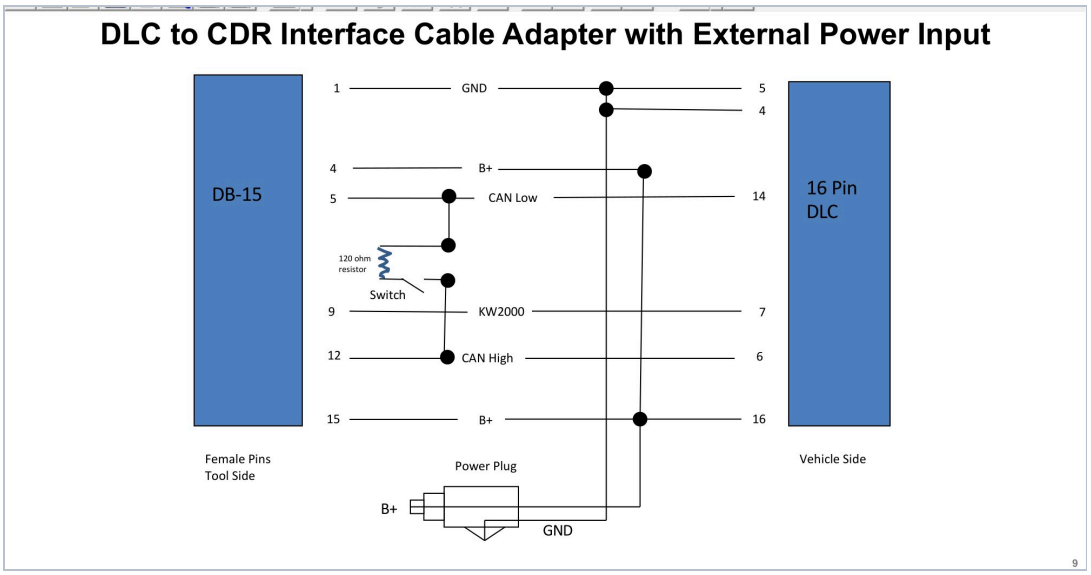
Vehicle Connections

Toyota provides **no direct-to-module connection** for Vehicle Control History. The retrieval must be made through the vehicle's 16-pin Diagnostic Link Connector (DLC), normally under the dash on the driver's side. The VCH data itself is held in the airbag ECU, which Toyota calls the *Airbag Electronic Control Unit*.

Powering the Module

Because the retrieval runs through the DLC, the module must stay powered for the whole read. Where vehicle power is unavailable or unreliable, the module can be backpowered through the DLC in the same way as a CDR download. A read-out can take anywhere from one to ninety minutes, so provide a booster or supply battery rather than relying on a damaged vehicle's own battery.

> **Losing power mid-read will end the retrieval.** Confirm your power source before pressing Next on the read-out screen, not after.



DLC to interface cable adapter with external power input — the arrangement used to backpower the vehicle through the 16-pin DLC during a retrieval.

About Techstream and Vehicle Control History

Vehicle Control History is a separate recording from the airbag EDR record read by the Bosch CDR tool. Retrieving one is not a substitute for retrieving the other.

What VCH Records

A CDR download alone may miss data that VCH holds. VCH records **post-trigger** data as well as pre-trigger, which makes it valuable in non-deployment events, multiple impacts, and sideswipes that the CDR report does not capture.

VCH includes the familiar parameters — vehicle speed, engine RPM, accelerator position, brake switch — and adds yaw rate, lateral and longitudinal G sensor signals, brake oil pressure, steering angle, shift position, and the ADAS system parameters. Events may also carry forward-facing camera images.

At least 26 separate trigger types cause an event record to be written. Some events record at 2 Hz and others at 6.67 Hz; most triggers capture roughly five seconds before and after the trigger. The authoritative trigger list is in the Vehicle Control History section of the online service manual at techinfo.toyota.com, which requires a subscription.

Covered Vehicles

Vehicle Control History was introduced progressively:

- 2013 — Toyota RAV4
- 2014 — Toyota Highlander; Lexus IS250/350
- 2015 — Toyota Camry, Corolla, Prius C, Prius V; Lexus NX and RC
- 2016 — Toyota Avalon, Land Cruiser, Mirai, Prius, Tacoma; Scion iM; Lexus ES, GS, LX and RX
- Later model years are generally supported; confirm coverage for the specific vehicle before you rely on it.

TSS 1.0 vs. TSS 2.0

The generation of Toyota Safety Sense on the vehicle changes what images you can recover and how many boxes appear on the image-selection screen.

- **TSS 1.0** — images are stored only for PCS events, and only if *the vehicle itself braked*. A PCS warning buzzer alone is not enough to link and store images. The retrieval screen offers a single choice, *PCS related Image data*.
- **TSS 2.0** (2019 and later RAV4, Corolla hatchback, Lexus ES350 and onward) — image capture no longer requires the vehicle to brake, and images may be stored against PCS, LKA, EDR, ICS and vehicle-behavior triggers. The retrieval screen offers five separate checkboxes.

Overwrite and Evidence Loss

> **VCH records are overwritten.** Events are sorted into recording “Areas,” and each Area holds a limited number of records before the oldest is replaced — the pre-collision Area holds about ten records, the lane-departure Area about twelve. Because of the overwrite rules the file numbers are not necessarily chronological.
>
> **Every key cycle risks losing data.** Retrieve VCH as early in the inspection as practical, and do not drive or repeatedly cycle the vehicle beforehand.

Equipment and Subscription

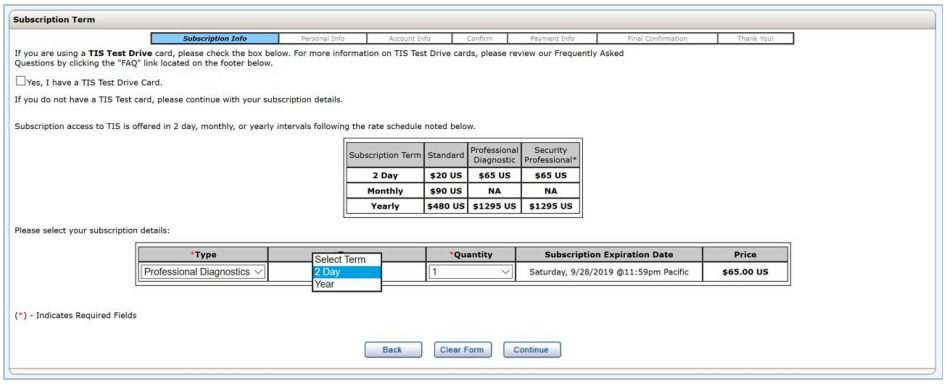
Hardware

- A Windows laptop meeting the Techstream requirements, with a Java runtime installed.
- A vehicle interface module (VIM) with a 16-pin DLC connector. Commonly used devices include the Bosch CDR 900, the Toyota MongoosePro MFC or MFC2, and the USB-Link 2.
- A booster or supply battery and, if you intend to backpower, a DLC adapter cable with an external power input.
- External media — a USB drive — to carry the retrieval files off the laptop before you leave the scene.

TIS Subscription

Techstream runs on a subscription from `techinfo.toyota.com`. Register at `techinfo.snapon.com/TIS/Register.aspx` if you do not already have an account.

> **Select “Professional Diagnostics,” not “Standard.”** A Standard subscription will not run Techstream. Subscriptions are sold in 2-day, monthly and yearly terms; a 2-day term is enough for a single inspection but see the warning below.



Subscription Term screen — set Type to Professional Diagnostics, choose the term, and set Quantity to 1.

Activation is not automatic. TIS emails an activation link with “Action Required” in the subject line, and the subscription stays inactive until you click it. That email frequently lands in a spam folder — check there before assuming the order failed.

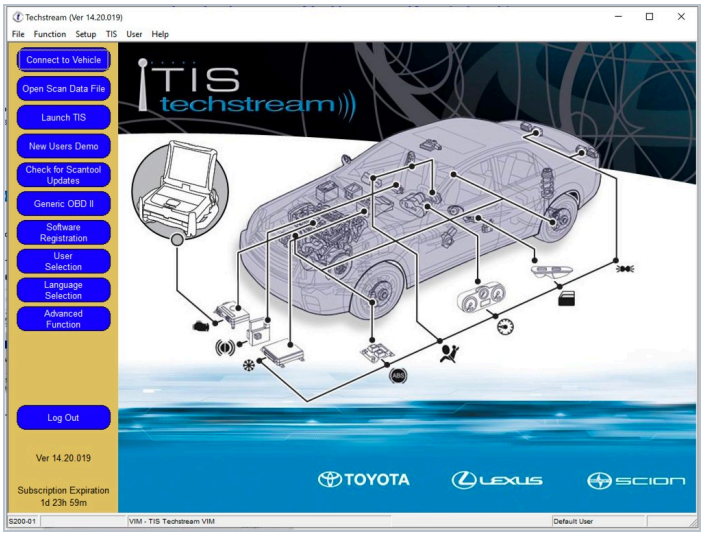
Software Installation

Log in, accept the terms of use, and download the current **Full** Techstream installer from the TIS tab under **Diagnostics**. Save the installer rather than running it directly so a reinstall is possible later. Select English (United States) and let the installer run; it takes a while. Set the region to North America and the user type to Public User, completing every field on that screen. If prompted about Java, install it.

On first launch Techstream asks for a registration key. Click **Software Registration**, log in, enter a Computer Name and Shop Owner, click **Calculate Key**, then copy the key back into the **New Key** field and click OK.

Confirm the Subscription Before You Travel

The Techstream main screen shows the software version and the remaining subscription time in the lower left. Confirm it reads the expected term before you leave for the inspection.

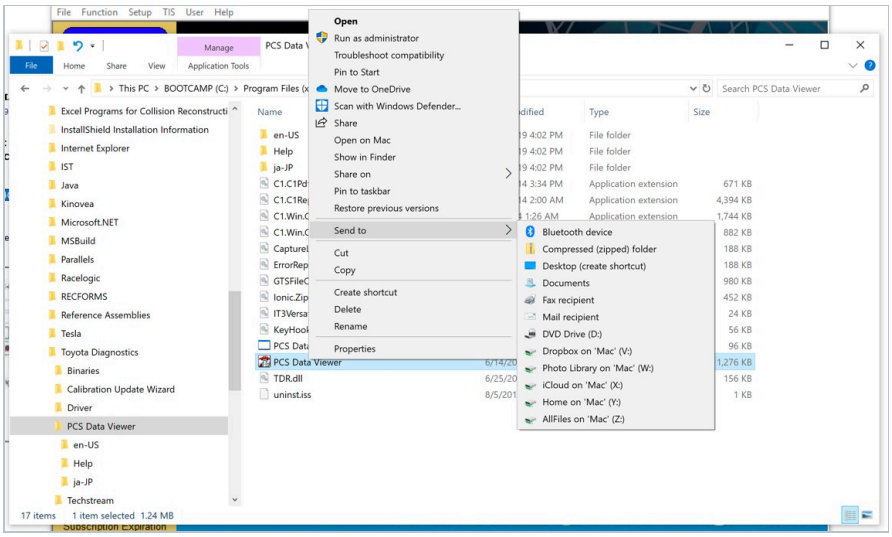


Subscription Expiration on the main screen. A 2-day term reads in hours and minutes.

> **A 2-day licence stops you reading your own evidence.** Once the licence expires you can no longer open the retrieved `.vdas` file in PCS Data Viewer. Complete the PCS Data Viewer extraction in Part 2 — the CSV, image and PDF output — *while the subscription is still live*, or buy a term long enough to cover the analysis.

Confirm PCS Data Viewer Is Installed

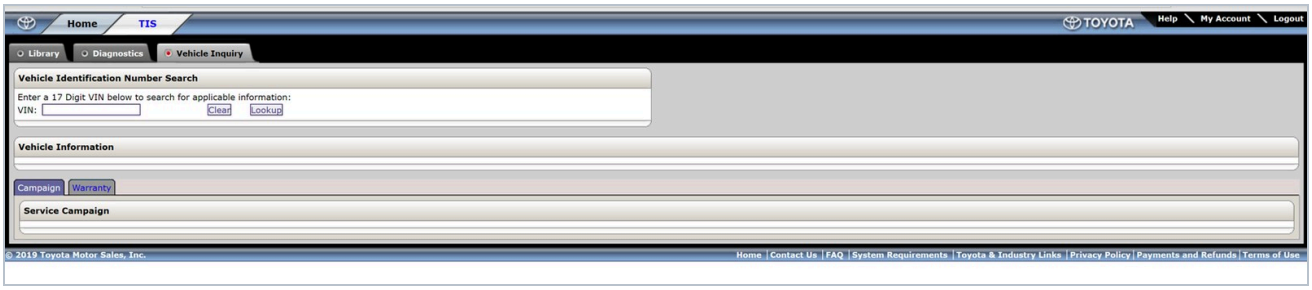
PCS Data Viewer is the application that opens the retrieval file. It installs alongside Techstream, in the Techstream program folder. Confirm it is present and, if there is no desktop shortcut, create one before you go.



PCS Data Viewer in the program folder — send a shortcut to the desktop.

Optional: Pull the As-Built Record First

The same subscription gives access to service manuals and the vehicle's factory "as-built" equipment record. Entering the VIN under the **Vehicle Inquiry** tab and expanding **Standard Equipment** tells you what safety systems the vehicle actually left the factory with — useful for knowing what to expect before you connect, and printable for the file.



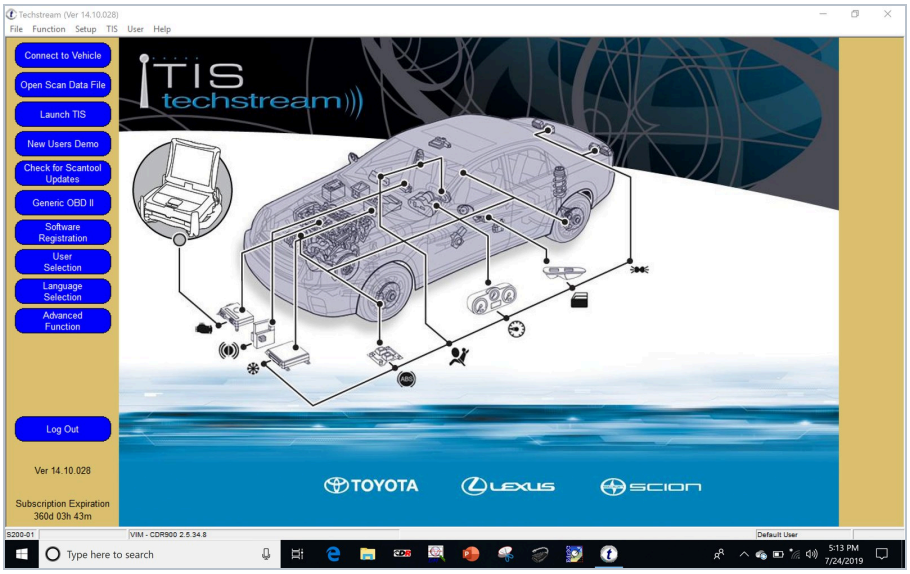
Vehicle Inquiry — enter the 17-digit VIN and click Lookup.

Imaging Data Using Techstream

Screen-capture every screen listed below as you go. The screen captures are the record of how the retrieval was performed and of the tool state at the time; they cannot be recreated later.

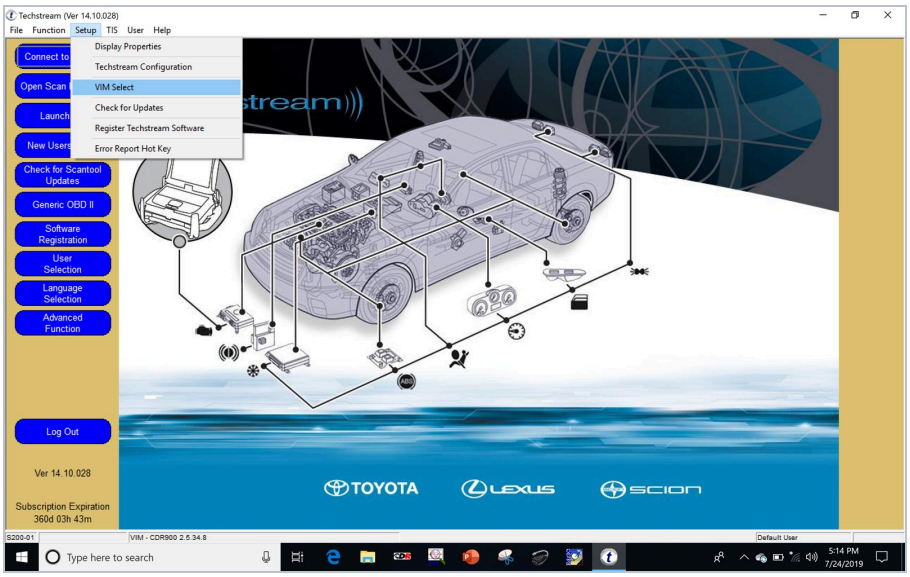
1. Verify system power by checking for 12 volts at the vehicle's battery. Turn the ignition **ON** — do not start the engine. Connect the VIM to the DLC and to the laptop, and confirm the laptop is booted and stable before opening Techstream.

2. Launch Techstream. Once it opens, **take a screen capture of the main screen**, which shows the software version and the remaining subscription time in the lower left, and the selected VIM in the status bar along the bottom.



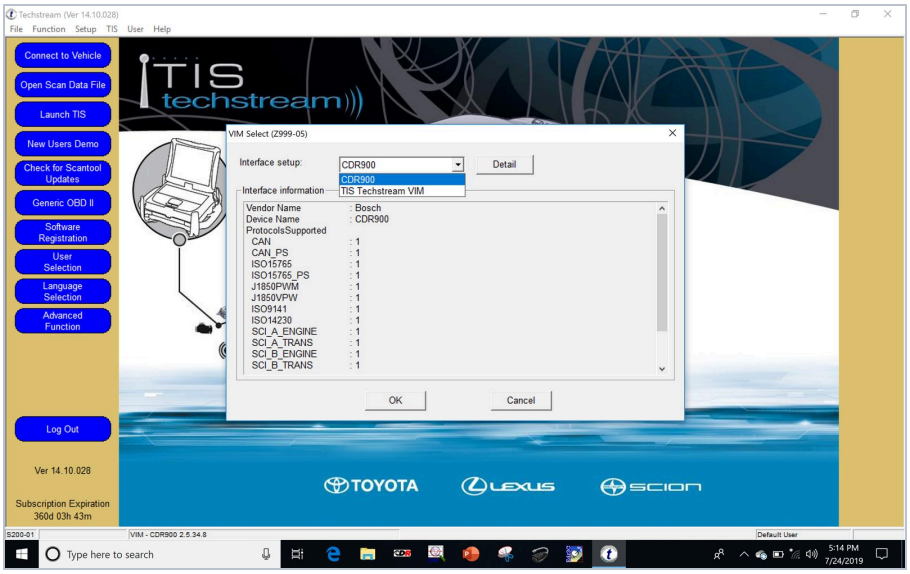
Step 2 — Techstream main screen. Version, Subscription Expiration and the active VIM are all visible; capture this screen.

3 Click **Setup**, then **VIM Select**.



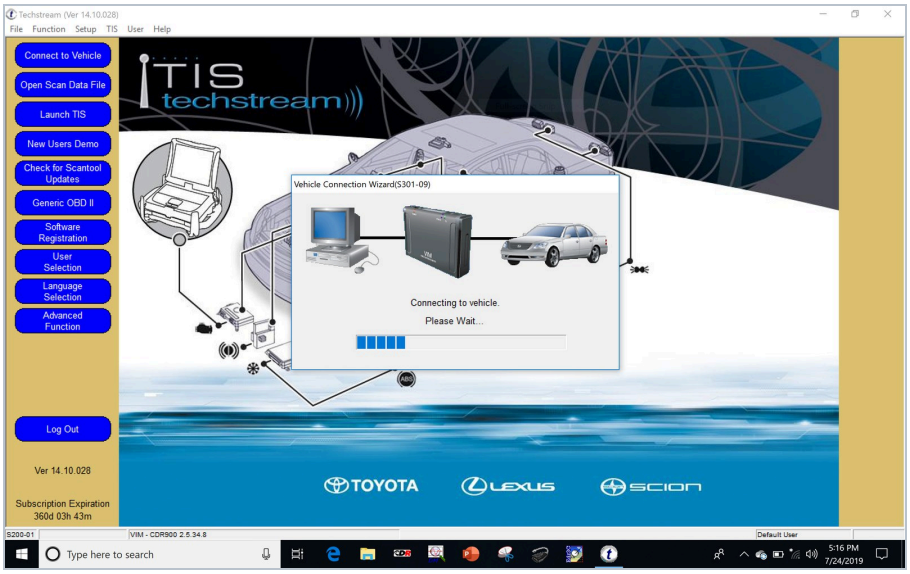
Step 3 — Setup → VIM Select.

4 Choose the interface you are using from the **Interface setup** list — several may be listed — and click **OK**. Take a screen capture showing the device selected and its reported vendor, device name and supported protocols.



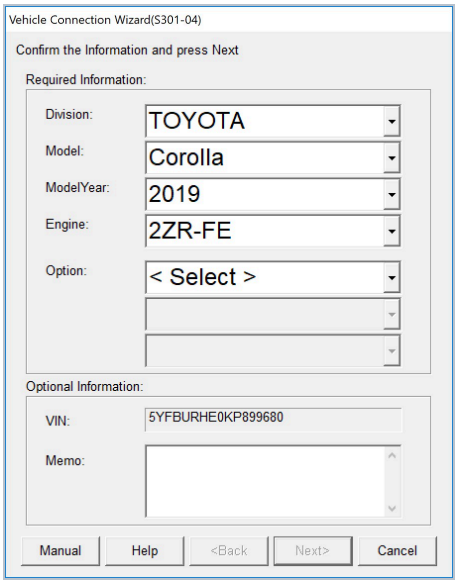
Step 4 — VIM Select. The interface information panel confirms which device Techstream will talk through.

5 Click **Connect to Vehicle** on the left of the main screen. Confirm the ignition is still ON. The Vehicle Connection Wizard reports its progress as it connects.



Step 5 — the Vehicle Connection Wizard connecting to the vehicle.

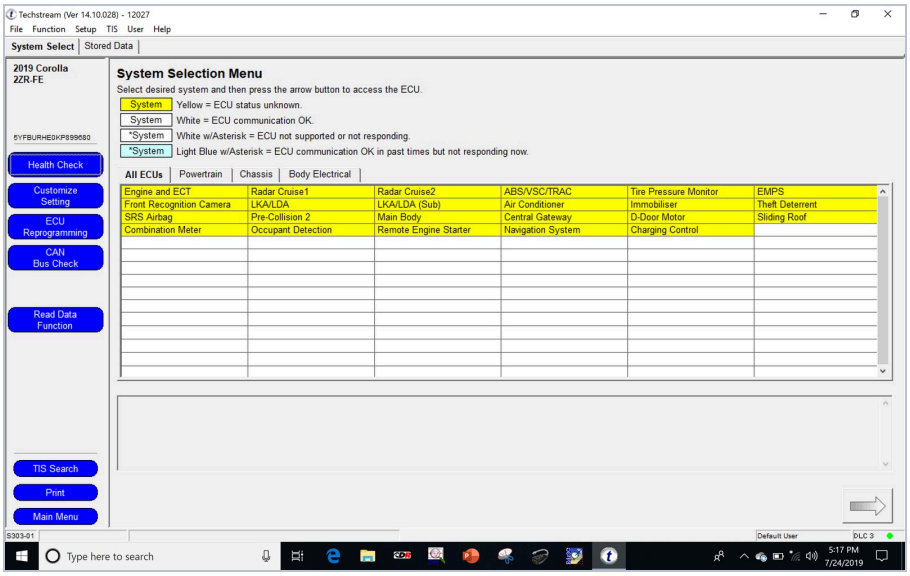
6 Techstream normally auto-detects the vehicle and populates the VIN. Confirm Division, Model, Model Year and Engine, set any options the vehicle carries (Smart Key, Radar Cruise), add a note in the Memo field identifying the case, and click **Next**. **Screen-capture this screen** — it is your record that the tool was connected to this VIN.



Step 6 — confirm the vehicle information and VIN.

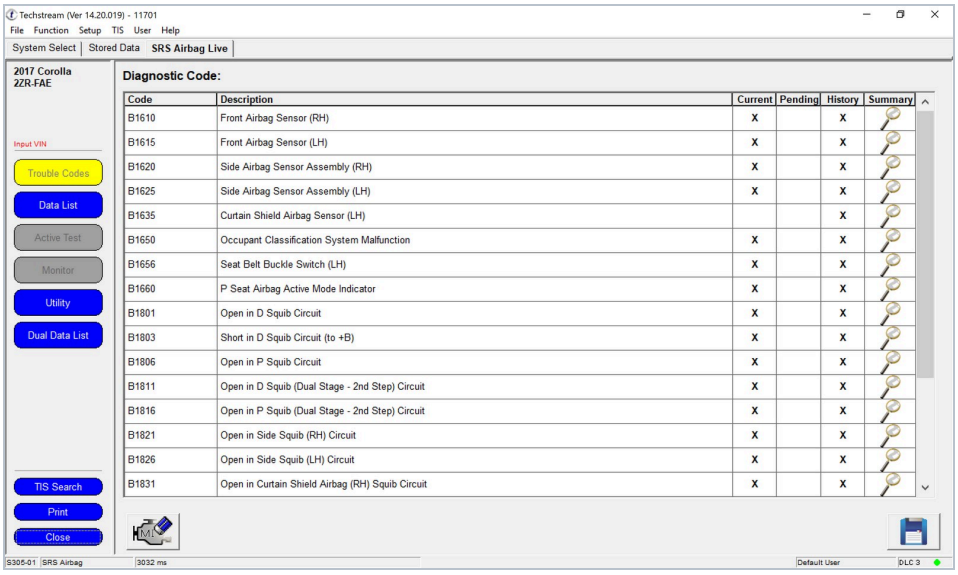
> **If you connect manually (direct-to-module / D2M), you must type the VIN yourself.** Techstream builds the retrieval file name from the VIN, and a blank or wrong VIN produces a VIN mismatch error and a misnamed file.

7 The System Selection Menu appears. Yellow cells are possible equipment whose status is unknown; running **Health Check** performs a bus scan and turns the ECUs that actually responded white. Run the Health Check, save or print its report, and **screen-capture the System Selection Menu** as your record of the ECU inventory on this vehicle.



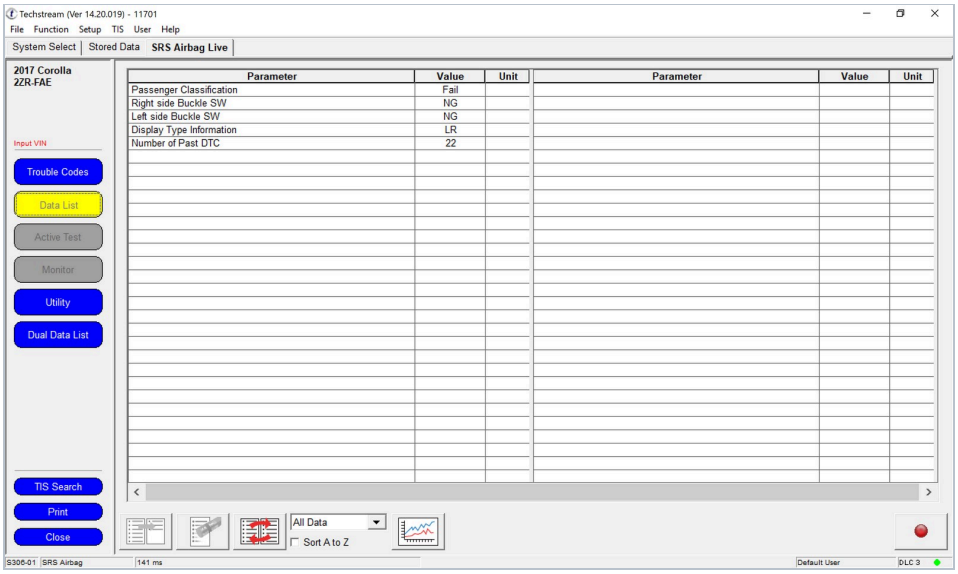
Step 7 — System Selection Menu. The colour key at the top explains ECU status.

8 Select **SRS Airbag** (wording varies by vehicle) and click **Trouble Codes**. Record the current, pending and history codes — save them with the disk icon at the lower right and take a screen capture.



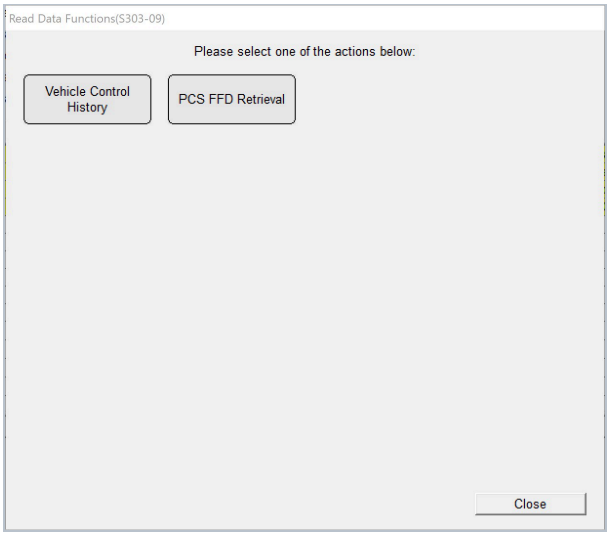
Step 8 — SRS Airbag diagnostic codes, showing Current, Pending and History columns.

9 Click **Data List** for the additional airbag-system information — occupant classification, buckle switches, number of past DTCs. Save and capture this screen as well.



Step 9 — SRS Airbag Data List.

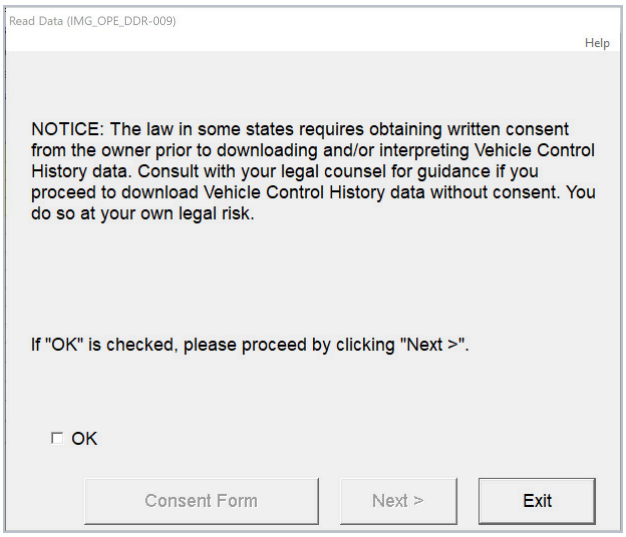
10 Return to the **System Select** tab and click **Read Data Function**. Two paths are offered. Choose **PCS FFD Retrieval**.



Step 10 — Read Data Functions. Choose PCS FFD Retrieval.

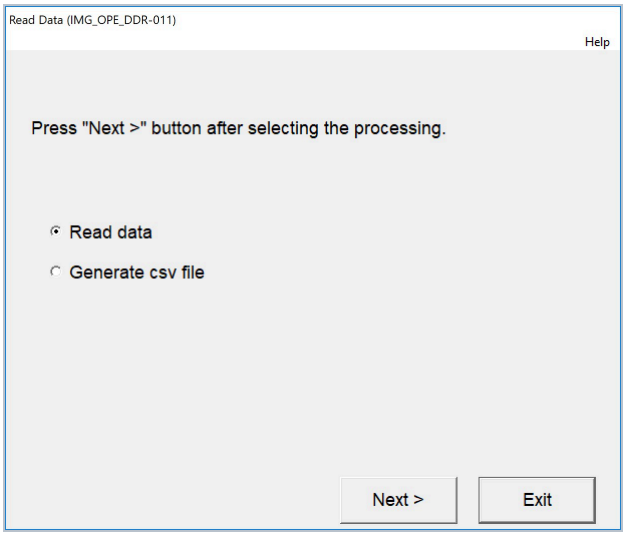
> **Why not the Vehicle Control History button?** The Vehicle Control History path retrieves VCH only. **PCS FFD Retrieval retrieves the Vehicle Control History *and* the camera images**, if any are present. Always take the PCS FFD Retrieval path so nothing is left behind. The same prompts can also be reached from the SRS Airbag section under **Utility**, but that route is not recommended for the same reason.

11 A legal notice appears. Some states require the owner's written consent before Vehicle Control History data is downloaded or interpreted. Satisfy yourself that you have the authority to proceed, **tick the OK checkbox**, and click **Next**. Screen-capture this screen.



Step 11 — the consent notice. Next stays greyed out until OK is ticked.

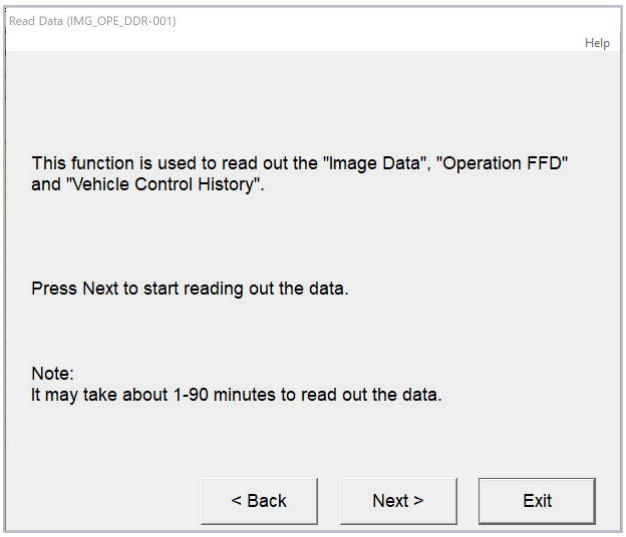
12 Select **Read data** and click **Next**.



Step 12 — choose Read data.

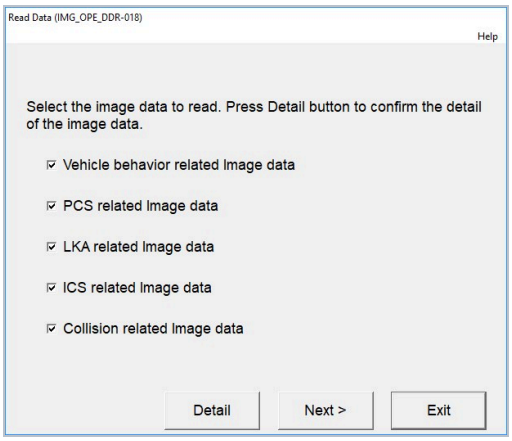
> **Do not choose “Generate csv file” at the vehicle.** That option does not retrieve the image data. Read the full data set first; the CSV output is produced later from the saved file in PCS Data Viewer.

13 Techstream confirms it is about to read out the Image Data, Operation FFD and Vehicle Control History, and warns that the read may take between one and ninety minutes. Confirm your power supply will last, then click **Next**.

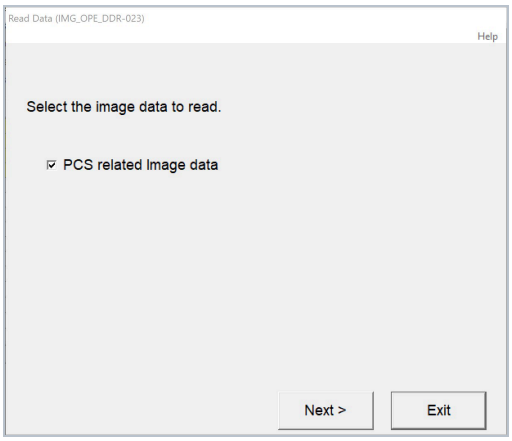


Step 13 — read-out confirmation and the 1–90 minute warning.

14 Select the image data to read, and **check every box offered**. A TSS 1.0 vehicle offers only *PCS related Image data*; a TSS 2.0 vehicle offers five categories. There is no reason to leave any unchecked. **Screen-capture this screen** — it records which categories were requested. Click **Next**.

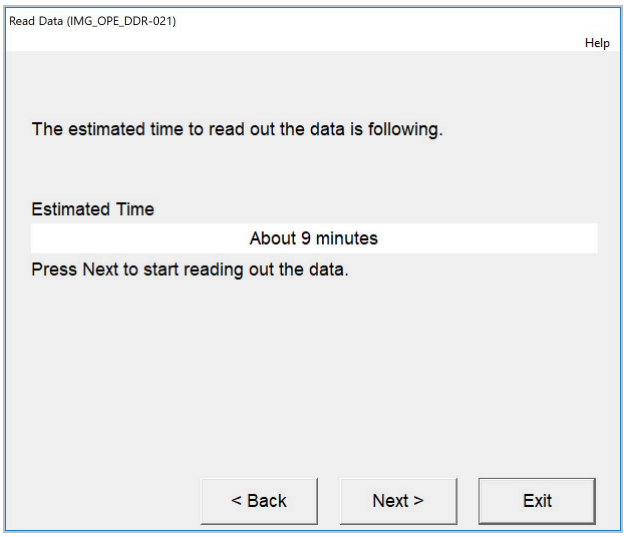


Step 14 — TSS 2.0 image selection: vehicle behavior, PCS, LKA, ICS and collision related image data. PCS = Pre-Collision System, LKA = Lane Keeping Assist, ICS = Intelligent Clearance Sonar.



Step 14, TSS 1.0 — a single choice, PCS related Image data.

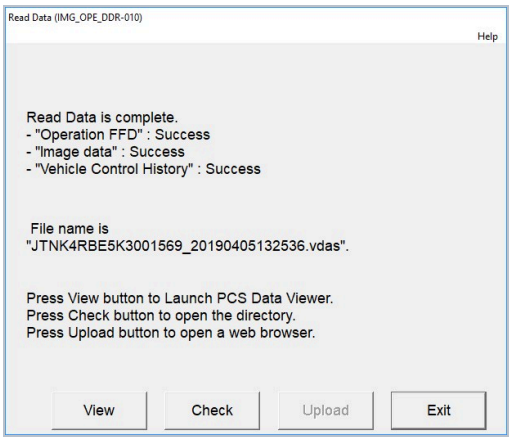
15 Techstream estimates the read time from the number of stored events. Confirm the estimate against your power supply and available time, then click **Next** to begin the read. Do not disturb the laptop or the connection while it runs.



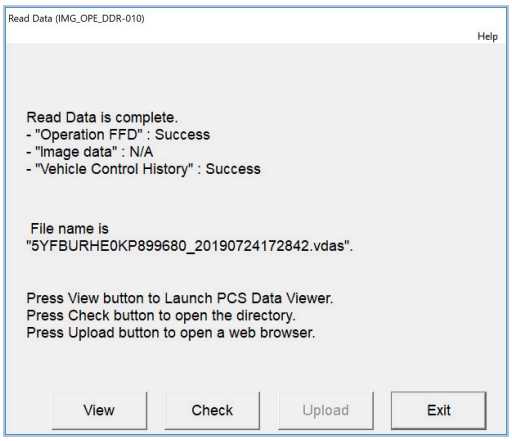
Step 15 — estimated read time for this vehicle.

16 When the read completes, Techstream reports the status of each of the three data sets and names the file it created. **Screen-capture this screen. It is the single most important capture of the retrieval** — it is your record of the file name and of whether each data set succeeded.

Read the three status lines carefully. `Image data : N/A` means no images were stored for the events retrieved, which is a finding in itself — not a failure of the retrieval.

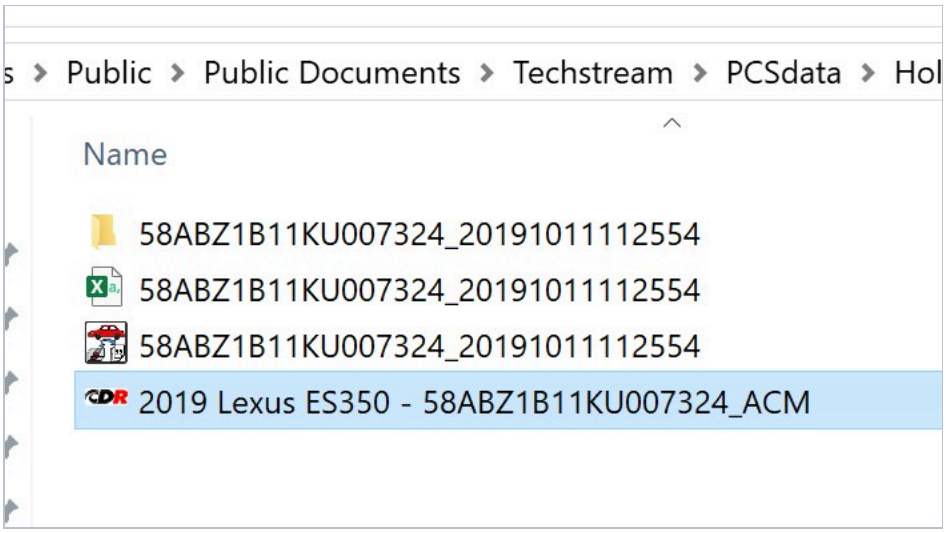


Step 16 — all three data sets successful. The file name follows the pattern `VIN_YYYYMMDDHHMMSS.vdas`.



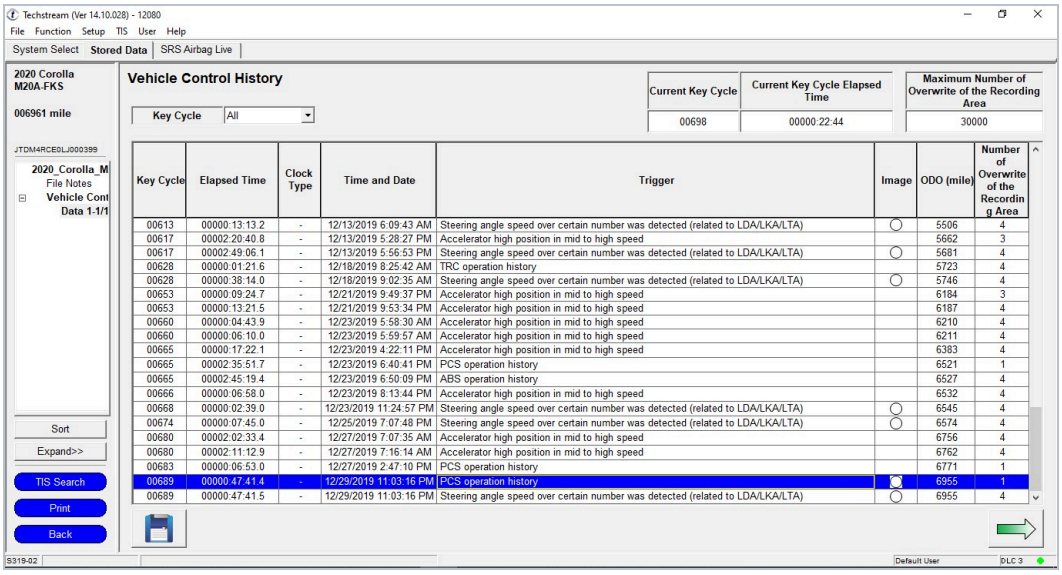
Step 16 — the same screen where no images were stored: Image data reads N/A while Operation FFD and Vehicle Control History both succeeded.

17 Click **Check**. Windows Explorer opens at the folder holding the retrieval file, by default `C:\Users\Public\Public Documents\Techstream\PCSdata\`. Confirm the `.vdas` file is present and note the full path.



Step 17 — the saved retrieval file in the PCSdata folder.

18 While still on scene and still licensed, you may review the events in Techstream itself: the **Stored Data** tab lists the Vehicle Control History with key cycle, elapsed time, date and time, trigger description, odometer and overwrite count. **A circle in the Image column marks an event that has camera images.** Identify the events relevant to your investigation and note their key cycles.



Step 18 — the VCH event list in Techstream. The disk icon at the lower left exports the list.

19 Opening an individual event shows the recorded parameters for that trigger and a trigger note. The playback controls step through each data point. Use the **disk icon** to export the event to Excel.



Step 19 — an individual trigger event with its parameter list and trigger note.

> **Exporting to Excel does not export the camera images.** The images come out only through PCS FFD Retrieval and the PCS Data Viewer output in Part 2. If an event carries images, you must complete Part 2 to preserve them.

20 **Copy the `.vdas` file and every screen capture to external media before you disconnect.** Work through Table 1 opposite and confirm each item is in hand. When everything is accounted for, exit Techstream, disconnect the VIM from the DLC, and turn the ignition off.

Table 1 — Files You Should Have Before Leaving the Inspection

At this point the following should have been imaged from the vehicle. Do not disconnect until every row is accounted for. Where a row is marked *not applicable*, record *why* in your notes rather than leaving it blank.

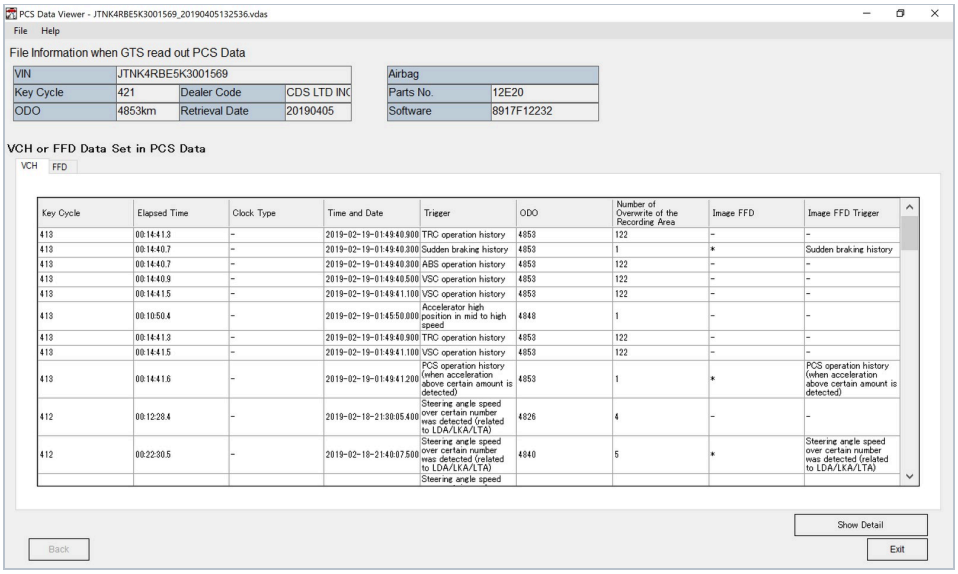
No.	File	Location	Save Method
1	Software version and subscription expiration	Techstream main screen at launch	Screen capture
2	Interface (VIM) selection and protocols	Setup → VIM Select	Screen capture
3	Vehicle identification and VIN	Vehicle Connection Wizard	Screen capture
4	ECU inventory / System Selection Menu	System Select tab	Screen capture
5	Health Check report	System Select → Health Check	Save / print to PDF
6	SRS Airbag diagnostic trouble codes	SRS Airbag → Trouble Codes	Save (disk icon) + screen capture
7	SRS Airbag data list	SRS Airbag → Data List	Save (disk icon) + screen capture
8	Consent / legal notice as acknowledged	Read Data (IMG_OPE_DDR-009)	Screen capture
9	Image data categories requested	Read Data image selection screen	Screen capture
10	Read-out estimated time	Read Data (IMG_OPE_DDR-021)	Screen capture
11	**Read Data complete summary** — status of Operation FFD, Image data and Vehicle Control History, plus the retrieval file name	Read Data (IMG_OPE_DDR-010)	Screen capture
12	**Retrieval file** `VIN_YYYYMMDDHHMMSS.vdas`	`C:\Users\Public\Public Documents\Techstream\PCSdata\`	Copy to external media
13	Vehicle Control History event list	Stored Data → Vehicle Control History	Export (disk icon) to Excel + screen capture
14	Individual trigger event data, for each event relevant to the investigation	VCH list → event → disk icon	Export to Excel + screen capture
15	Photographs of vehicle, VIN plate, odometer and DLC; laptop clock vs. actual time	Scene	Camera / notes
16	Vehicle as-built equipment record (optional, from TIS)	techinfo.toyota.com → Vehicle Inquiry	Print to PDF

> **The `.vdas` file is the evidence.** Everything in Part 2 is produced from it. Verify it copied successfully to external media — open the copy, check the file size — before the laptop leaves the vehicle.

Extracting the Data Using PCS Data Viewer

Complete this part while the TIS subscription is still active. Once it expires, the `.vdas` file can no longer be opened.

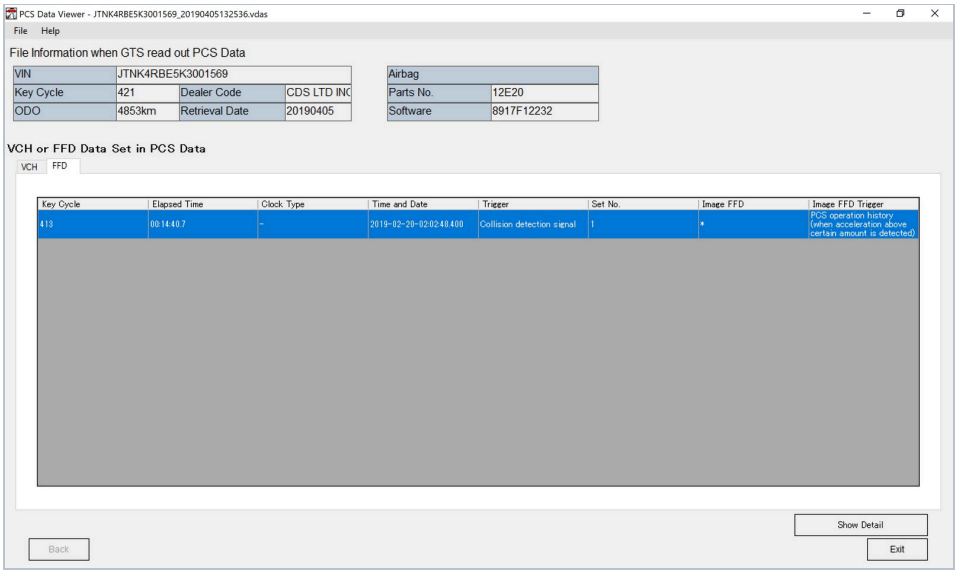
1. Launch **PCS Data Viewer** from the desktop shortcut. Click **File → Load VDAS File** and open the retrieval file. The header fills in with the VIN, key cycle, odometer, retrieval date, dealer code and the airbag ECU part number and software number.



Part 2, Step 1 — the VCH tab of a loaded retrieval file. Double-click the Key Cycle header to sort highest to lowest.

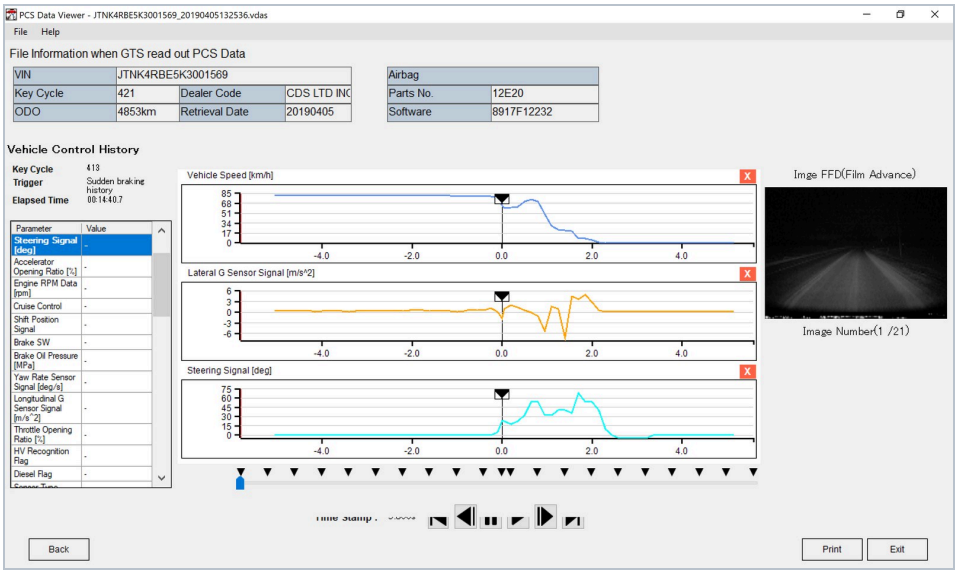
2. Work the **VCH** tab. Each row is one trigger event: key cycle, elapsed time, clock type, time and date, trigger description, odometer, overwrite count, and whether an Image FFD record exists. An asterisk in the **Image FFD** column marks an event with camera images.

3. Work the **FFD** tab in the same way. VCH and FFD views of the same event contain overlapping but not identical data — review both.



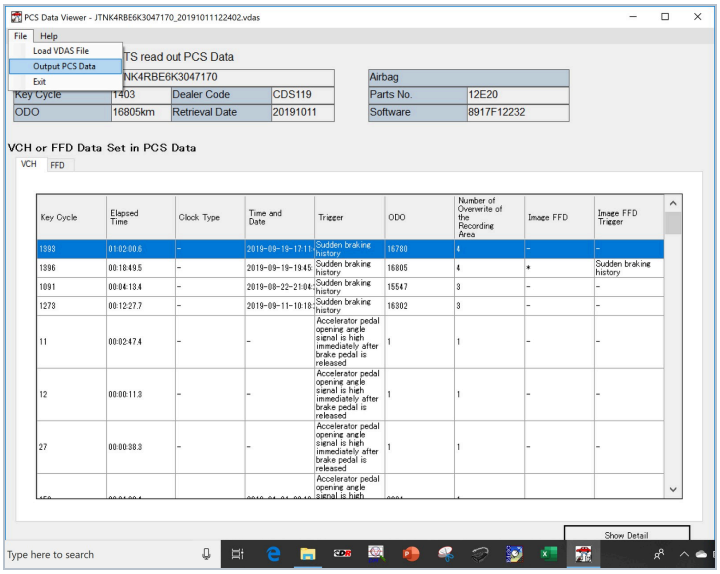
Part 2, Step 3 — the FFD tab, here holding a single collision detection record.

4. Double-click an event to open playback. Selected parameters are graphed against time with the trigger at zero. Where the event carries images, they load alongside the graphs and step in sync with the playback controls.



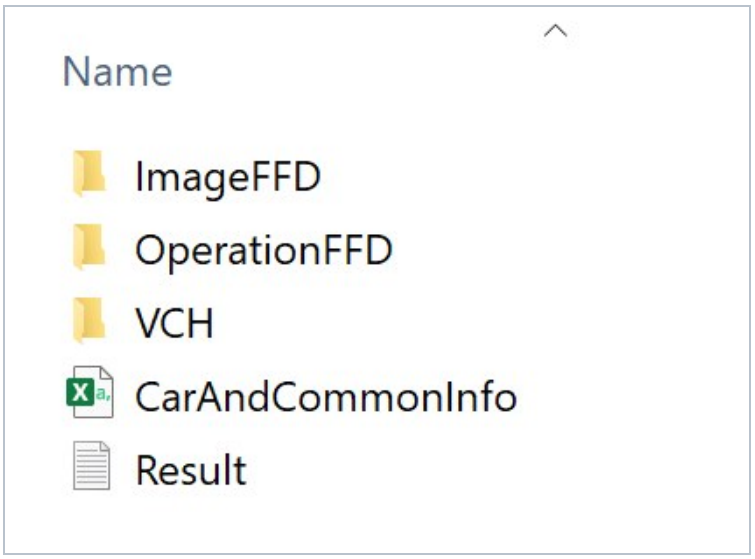
Part 2, Step 4 — event playback with vehicle speed, lateral G and steering plotted, and the synchronised forward-facing camera image at the right.

5 Click **File → Output PCS Data** and choose a destination. Save the output alongside the `.vdas` file so the retrieval and its extraction stay together.



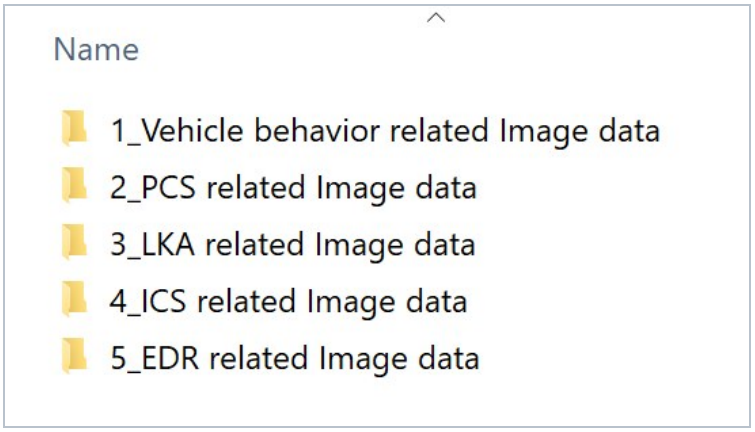
Part 2, Step 5 — File → Output PCS Data.

6 The output is written into three directories plus two files: **ImageFFD**, **OperationFFD** and **VCH**, with **CarAndCommonInfo** holding the vehicle and tool data and **Result** recording the output run. Confirm all five are present.

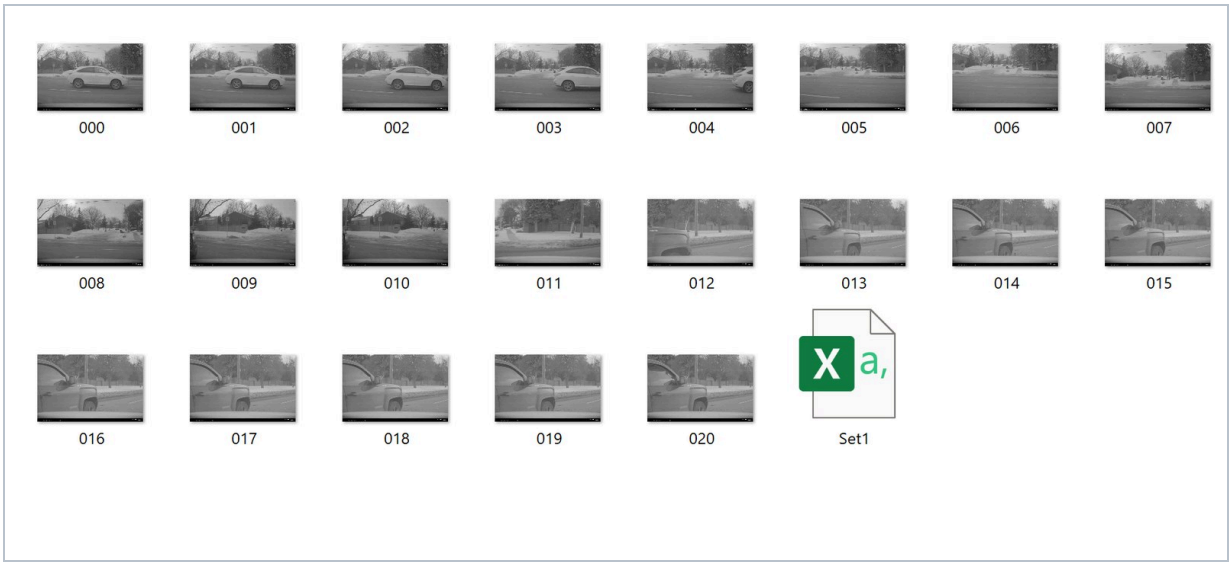


Part 2, Step 6 — the five output items.

7 On a TSS 2.0 vehicle the image output is split by trigger type into separate folders. Images may be duplicated across folders where triggers occurred concurrently — that is expected, not an error. Each category holds one to three sets of images, and each set is accompanied by a `Setx` CSV file linking the images to the data.

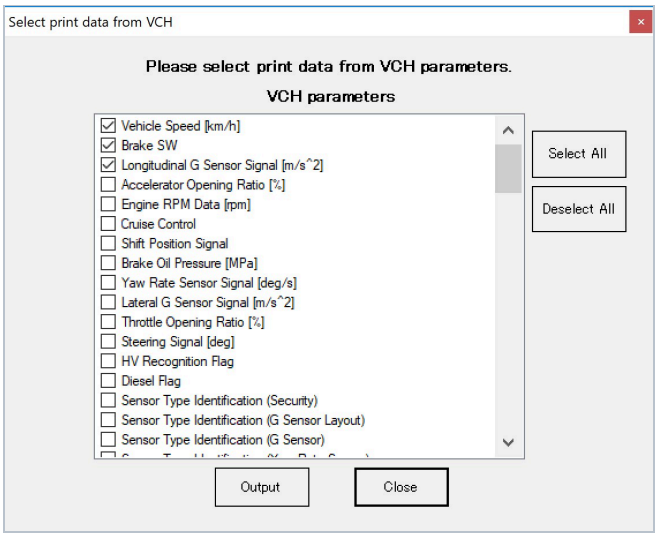


Part 2, Step 7 — image folders by trigger type on a TSS 2.0 vehicle.

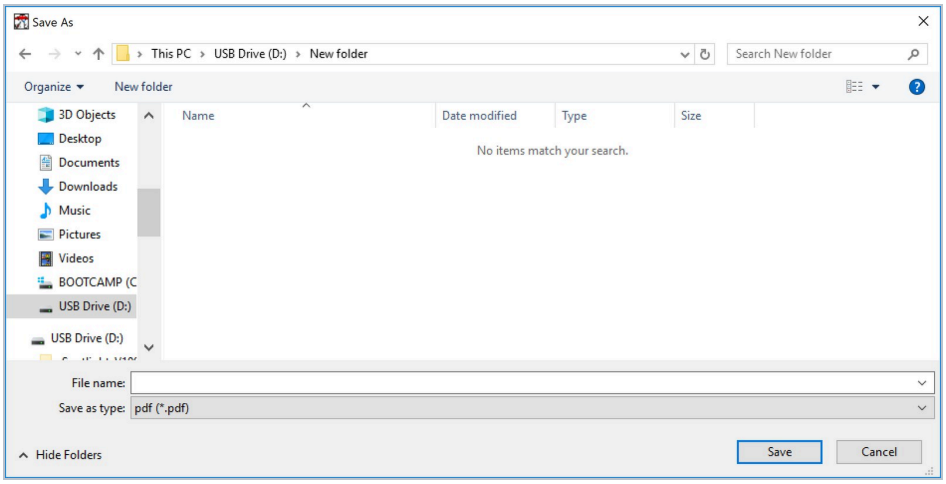


Part 2, Step 7 — extracted images, sequentially numbered, with the Set1 CSV that ties them to the event data. Images are low resolution, roughly 160 × 102 and monochrome, but often still probative.

8 Finally, produce a printed report. Click **Print**, select the parameters to include — **Select All** unless you have a reason not to — and click **Output**. Choose a destination and save as **PDF**. Any images are printed at the bottom of the report. You do not need anything loaded in the viewing area to print.



Part 2, Step 8 — parameter selection for the printed report.



Part 2, Step 8 — save the report as a PDF onto your case media.

Table 2 — Files Produced by PCS Data Viewer

Preserve the output below together with the original `.vdas` file. The CSV and image files remain readable after the subscription expires; the `.vdas` file does not.

No.	File	Location	Save Method
1	Vehicle Control History event data, one CSV per Area	`VCH\`	File → Output PCS Data
2	Operation Freeze Frame Data, one CSV per event	`OperationFFD\`	File → Output PCS Data
3	Forward-facing camera images, by trigger category	`ImageFFD\1_Vehicle behavior…5_EDR\`	File → Output PCS Data
4	Image set index files linking images to event data	`ImageFFD\…\Setx`	File → Output PCS Data
5	Vehicle and tool information	`CarAndCommonInfo`	File → Output PCS Data
6	Output result record	`Result`	File → Output PCS Data
7	Printed VCH / FFD report, images included	PCS Data Viewer → Print → Output	Save as PDF
8	Screen captures of the viewer showing each relevant event	PCS Data Viewer	Screen capture

Reading the Output

VCH events are sorted into recording **Areas**, each with its own overwrite rules and record capacity. Four Areas commonly relate to an EDR record:

- **Areas 1, 5 and 11** — brake control system events: sudden braking, traction control, vehicle stability control.
- **Area 6** — pre-collision detection, warning and auto-braking. Up to ten records before overwriting.
- **Area 8** — lane departure alert and lane keep assist. Up to twelve records before overwriting.

> **File numbers are not chronological.** Because of the overwrite rules, a lower Set number can hold a *higher* trip count than a higher Set number. Sort by trip count and key cycle, not by file name, and use the Time Count to link a VCH event to a CDR event.

A single retrieval can produce an Excel file with 300 or more data columns for each timestamp. Budget analysis time accordingly.

Source Material

This guide is a procedural compilation. The screenshots and the underlying procedure are the work of the authors below, and the guide should be read alongside their original material rather than in place of it.

- **Toyota TechStream / PCS & VCH Data**, Brad Muir, Crash Data Specialists LLC — CDR System Operators / Analysis & Applications Course Series, and the version edited and presented by Rick Ruth at the IPTM 2021 Symposium on Traffic Safety. Source of the setup, retrieval and PCS Data Viewer screenshots.
- **Toyota Techstream Software and Vehicle Control History Data — What Is It and How Do I Get It?**, E. Clarke, California Highway Patrol MAIT, September 2020. Source of the in-Techstream Vehicle Control History and trigger event screenshots, and of the licence and camera-image cautions.
- **Techstream Lite FAQ**, Toyota Motor Sales / techinfo.toyota.com, Rev 2.4.
- **SAE 2016-01-1495** — for the specific trigger list and recording rates.
- The Vehicle Control History section of the Toyota online service manual at techinfo.toyota.com, which is the authoritative reference for trigger definitions and parameters and requires a subscription.

Format modelled on the Cummins PowerSpec and Insite imaging guides by Cheek, Steiner, Plant and Austin.

> **Verify before you rely on it.** Techstream screens, menu wording and default file paths change between software versions. Confirm the current procedure against the service manual for the version you are running, and treat the screenshots here as representative rather than exact.